

□ * Basic concept of double integral, Double integral in cartesian and polar co-ordinate system.

* Concept of triple integral, Triple integral in Cartesian cylindrical and Spherical co-ordinate system.

* Introduction to differential equations, Separable variable, linear equations, Integrating factor.

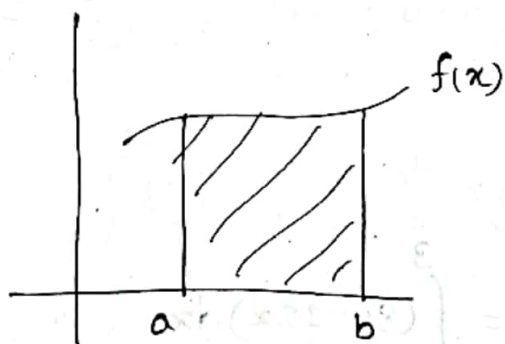
* Exact equation, non-exact equation, convert non-exact to exact.

* Higher order DE, homogeneous and non-homogeneous cases.

* modeling with 1st and second order DE.

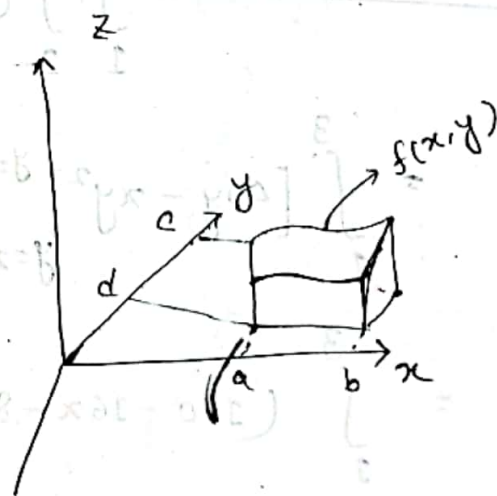
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□ Double integral:



area under a curve is obtained

by integral. $\int_a^b f(x) dx$.



Double integral gives volume under a curve or a surface

$$\int_c^d \left[\int_a^b f(x, y) dx \right] dy \quad \text{or,} \quad \int_a^b \left[\int_c^d f(x, y) dy \right] dx$$

Start with the innermost integral and work your way outwards.

Example : $f(x, y) = 1 + 4xy$

$$x = 0 \text{ to } x = 1$$

$$y = 1 \text{ to } y = 3$$

$$\begin{aligned} \int_1^3 \int_0^1 (1 + 4xy) dx dy &= \int_1^3 [x + 2x^2y]_0^1 dy \\ &= \int_1^3 (1 + 2y) dy \\ &= [y + y^2]_1^3 = 10 \end{aligned}$$

Example (02) : $\int_1^3 \int_2^4 (40 - 2xy) dy dx$

$$\begin{aligned} &= \int_1^3 [40y - xy^2]_{y=2}^{y=4} dx \\ &= \int_1^3 (160 - 16x - 80 + 4x) dx = \int_1^3 (80 - 12x) dx \\ &= [80x - 6x^2]_1^3 = 112. \end{aligned}$$

$$\int_2^4 \int_1^3 (40 - 2xy) dx dy$$

$$= 112 \text{ (same)}$$

Theorem (Fubini's Theorem)

Let R be the rectangle defined by the inequalities

$$a \leq x \leq b, \quad c \leq y \leq d$$

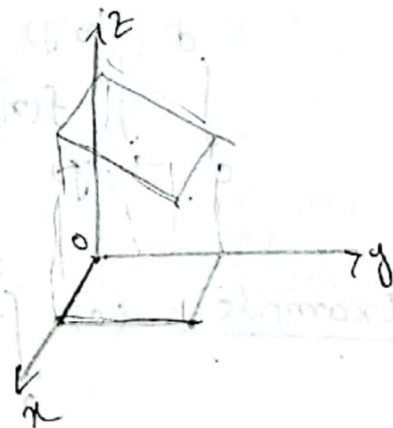
If $f(x, y)$ is continuous on the rectangle, then

$$\begin{aligned} \iint_R f(x, y) dA &= \int_c^d \int_a^b f(x, y) dx dy \\ &= \int_a^b \int_c^d f(x, y) dy dx. \end{aligned}$$

Example: Use a double integral to find the volume of the solid that is bounded above by the plane $z = 4 - x - y$ and below by the rectangle $R = [0, 1] \times [0, 2]$.

Solⁿ:

$$\begin{aligned} \int_0^2 \int_0^1 (4 - x - y) dx dy &= V \\ &= \int_0^2 (4x - x^2 - xy)_0^1 dy \\ &= \int_0^2 \left(\frac{7}{2} - y \right) dy \\ &= \left[\frac{7}{2}y - \frac{y^2}{2} \right]_0^2 = 5. \end{aligned}$$



Example: Evaluate the double integral,

$$\iint_R y^2 x \, dA$$

over the rectangle $R = \{(x, y) : -3 \leq x \leq 2, 0 \leq y \leq 1\}$.

Solⁿ: $\int_0^1 \int_{-3}^2 (y^2 x) \, dx \, dy$

$$= \int_0^1 \left[\frac{x^2}{2} y^2 \right]_{-3}^2 dy = \int_0^1 \left(-\frac{5}{2} y^2 \right) dy = -\frac{5}{2} \left[\frac{y^3}{3} \right]_0^1$$

$$= -\frac{5}{6}.$$

Practice: 14.1 $\rightarrow$ 1-16, 29-31

▣ Double integrals over nonrectangular regions:

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \right] dx$$

$$\int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \right] dy$$

Example: (a) $\int_0^1 \int_{-x}^{x^2} y^2 x \, dy \, dx$

$$= \int_0^1 \left[\frac{y^3}{3} x \right]_{-x}^{x^2} dx = \int_0^1 \left(\frac{x^7}{3} + \frac{x^4}{3} \right) dx$$

$$= \left[\frac{x^8}{24} + \frac{x^5}{15} \right]_0^1$$

$$= \frac{13}{120}.$$

(b) $\int_0^{\pi/3} \int_0^{\cos y} x \sin y \, dx \, dy$

$$= \int_0^{\pi/3} \left[\frac{x^2}{2} \sin y \right]_0^{\cos y} dy = \int_0^{\pi/3} \left(\frac{1}{2} \cos^2 y \sin y \right) dy$$

$$= \frac{1}{2} \int_0^{\pi/3} u^2 dy$$

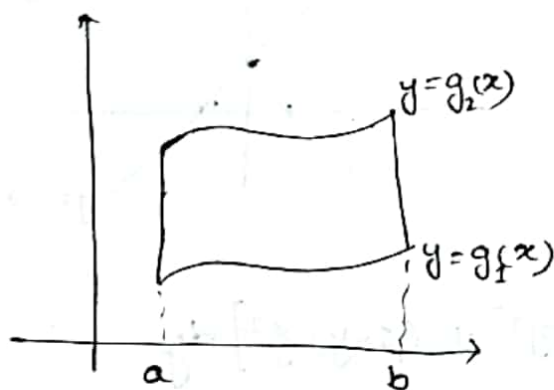
let $u = \cos y$

$$\Rightarrow du = -\sin y \, dy$$

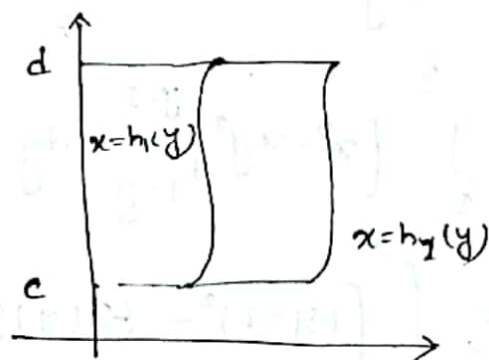
$$= -\frac{1}{2 \times 3} [u^3]_0^{\pi/3}$$

$$= -\frac{1}{6} [\cos^3 y]_0^{\pi/3} = \frac{7}{48}.$$

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$$\iint_R f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

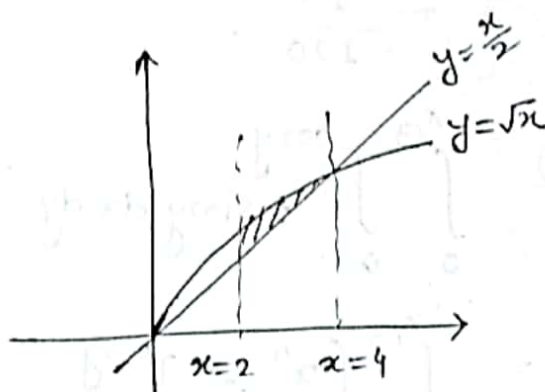


$$\iint_R f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

Example : Evaluate $\iint_R xy \, dA$

over the region bounded by the curves $y = \frac{1}{2}x$ & $y = \sqrt{x}$
 $x=2$ and $x=4$.

Solⁿ :
$$\int_2^4 \int_{x/2}^{\sqrt{x}} xy \, dy \, dx$$
$$= \frac{11}{6}.$$



Example (4) : $\iint_R (2x - y^2) \, dA \rightarrow$ Evaluate

over the triangular region R enclosed between the lines $y = -x+1$, $y = x+1$ and $y=3$.

$$\int_1^3 \int_{1-y}^{y-1} (2x - y^2) \, dx \, dy$$

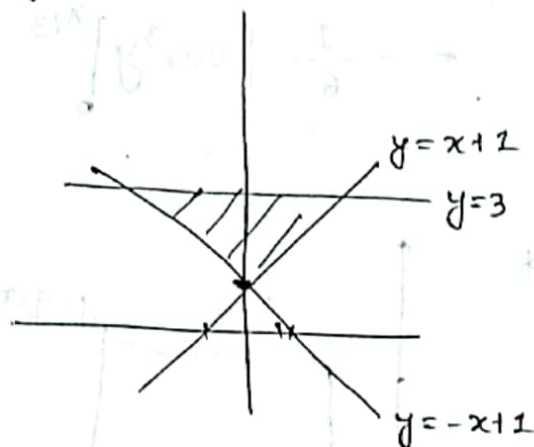
$$= \int_1^3 \left[x^2 - xy^2 \right]_{1-y}^{y-1} dy$$

$$= \int_1^3 \left[(y-1)^2 - (1+y)y^2 - (1-y)^2 + (1-y)y^2 \right] dy$$

$$= \int_1^3 (1 - 2y + y^2 + y^2 - y^3 - 1 + 2y - y^2 + y^2 - y^3) dy$$

$$= \int_1^3 (2y^2 - 2y^3) dy$$

$$= \left[\frac{2y^3}{3} - \frac{y^4}{2} \right]_1^3 = -\frac{68}{3}.$$

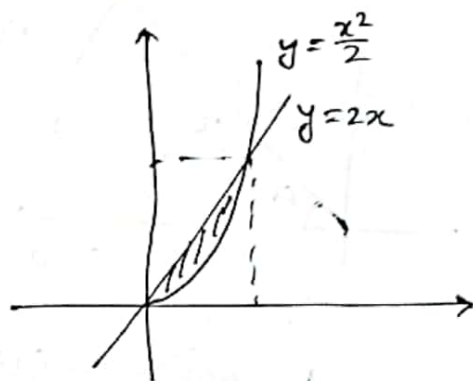


Example : Use a double integral to find the area of the region R enclosed between the parabola $y = \frac{x^2}{2}$ and the line $y = 2x$.

$$\text{Area of } R = \iint_R dA$$

$$= \int_0^4 \int_{x^2/2}^{2x} dy dx$$

$$= \int_0^4 \left[y \right]_{x^2/2}^{2x} dx = \int_0^4 \left(2x - \frac{x^2}{2} \right) dx = \left[x^2 - \frac{x^3}{6} \right]_0^4 = \frac{16}{3}.$$



Practice : 14.2 → 1-8, 14-23, 24

Cartesian to Polar co-ordinate:

Cartesian

x, y

dx, dy

$x = r \cos \theta$

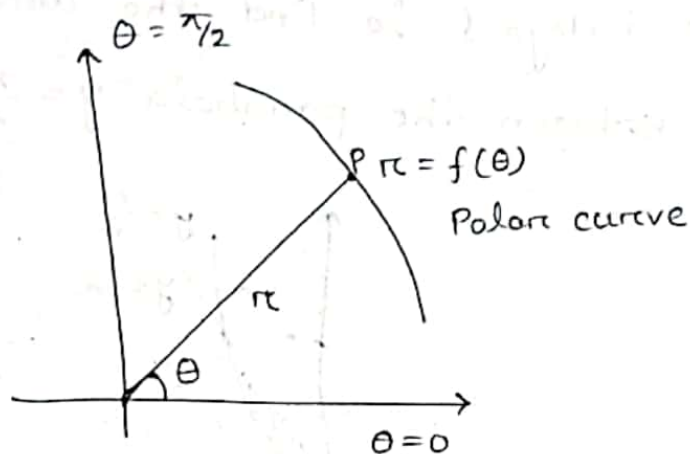
$y = r \sin \theta$

$x^2 + y^2 = r^2$

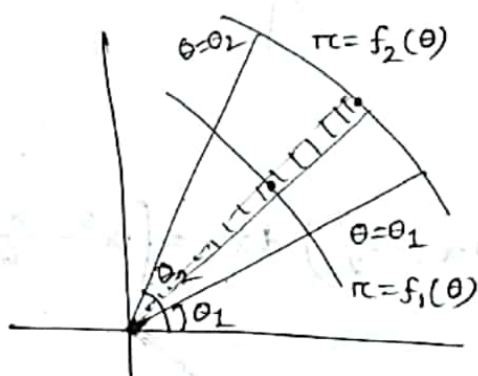
Polar

r, θ

$r dr, d\theta$



radius
 $r = \text{Polar vector}$
 $\theta = \text{vectorial angle}$
 origin $\rightarrow$ Pole



Then Double integral in Polar form is $= \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta) dr d\theta$

Example : Evaluate $\int_0^{\pi} \int_0^{a(1-\cos\theta)} r^2 \sin\theta dr d\theta$

$$= \int_0^{\pi} \left[\frac{r^3}{3} \sin\theta \right]_0^{a(1-\cos\theta)} d\theta$$

$$= \frac{a^3}{3} \int_0^{\pi} \sin\theta (1-\cos\theta)^3 d\theta$$

$$= \frac{a^3}{3} \int_0^2 (u)^3 du$$

$$= \frac{a^3}{3} \left[\frac{u^4}{4} \right]_0^2$$

$$= \frac{a^3}{3} \times \frac{2^4}{4}$$

$$= \frac{4a^3}{3} \text{ (Ans.)}$$

$$\text{let } u = 1 - \cos\theta$$

$$\Rightarrow du = \sin\theta d\theta$$

$$\text{when } \theta = 0, u = 0$$

$$\theta = \pi, u = 2$$

$$\cos 2\theta = 1 - 2\sin^2\theta$$

Example (2) $\int_0^\pi \int_0^{a\sin\theta} r \, dr \, d\theta$

$$= \int_0^\pi \left[\frac{r^2}{2} \right]_0^{a\sin\theta} d\theta = \frac{a^2}{2} \int_0^\pi \sin^2\theta \, d\theta = \frac{a^2}{2} \int_0^\pi \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

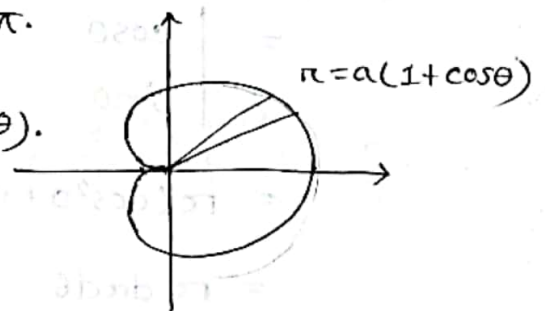
$$= \frac{a^2}{4} \int_0^\pi (1 - \cos 2\theta) \, d\theta = \frac{a^2}{4} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi$$

$$= \frac{a^2}{4} (\pi - 0) = \frac{a^2\pi}{4}$$

Example: Evaluate $\iint_A r \sin\theta \, dr \, d\theta$ over the area of the cardioid, $r = a(1 + \cos\theta)$ above the initial line.

Solⁿ: Here θ varies from 0 to π .

r varies from 0 to $a(1 + \cos\theta)$.



Now, $\int_0^\pi \int_0^{a(1+\cos\theta)} r \sin\theta \, dr \, d\theta$

$$= \int_0^\pi \sin\theta \left(\frac{r^2}{2} \right)_0^{a(1+\cos\theta)} d\theta$$

$$= \frac{a^2}{2} \int_0^\pi \sin\theta (1 + \cos\theta)^2 d\theta$$

$$= -\frac{a^2}{2} \int_2^1 u^2 du = \frac{a^2}{2} \int_1^2 u^2 du = \frac{a^2}{2} \left[\frac{u^3}{3} \right]_1^2 = \frac{4a^2}{3}$$

Let $1 + \cos\theta = u$

$\Rightarrow -\sin\theta \, d\theta = du$

when $\theta = 0$ $u = 2$

$\theta = \pi$ $u = 0$

Change of variable in double integral:-

Cartesian to polar form:

To transform cartesian coordinate (x, y) into polar coordinate (r, θ) , we define,

$$\begin{aligned} x &= r \cos \theta & \text{where } x^2 + y^2 &= r^2 \\ y &= r \sin \theta \end{aligned}$$

$$\therefore dx dy = J dr d\theta = \frac{\partial(x, y)}{\partial(r, \theta)} dr d\theta \quad J = \text{Jacobian}$$

$$= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} dr d\theta$$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} dr d\theta$$

$$= r(\cos^2 \theta + \sin^2 \theta) dr d\theta$$

$$= r dr d\theta$$

$$\Rightarrow dx dy = r dr d\theta$$

$$\text{Hence, } \iint_R f(x, y) dx dy = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta.$$

Steps : i) Given limits of integration and draw region.

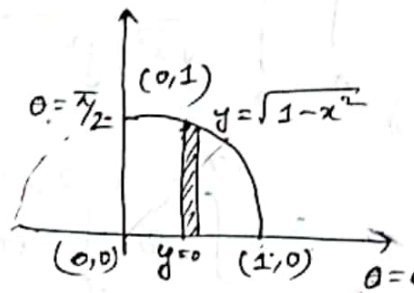
ii) put $x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$.

iii) Find the limit of r and θ .

iv) Solve.

Example: $\int_{x=0}^1 \int_{y=0}^{\sqrt{1-x^2}} dy dx$. Evaluate using change of variable in Polar form.

Solution: The limits of integration are, $\theta = \pi/2$



$x=0$ to $x=1$
 $y=0$ to $y = \sqrt{1-x^2} \Rightarrow x^2 + y^2 = 1$

Now, $x = r \cos \theta$ $y = r \sin \theta$ $dx dy = r dr d\theta$

So, we get,

$$\int_0^1 \int_0^{\sqrt{1-x^2}} dy dx = \int_0^{\pi/2} \int_0^1 r dr d\theta$$

$$= \int_0^{\pi/2} \left[\frac{r^2}{2} \right]_0^1 d\theta = \frac{1}{2} \int_0^{\pi/2} d\theta = \frac{1}{2} \left[\theta \right]_0^{\pi/2} = \frac{\pi}{4}$$

$x^2 + y^2 = 1$
 $\Rightarrow r^2 = 1$
 $\Rightarrow r = 1$

Example: Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$, by changing to Polar coordinate.

Solution: Limits of integration are

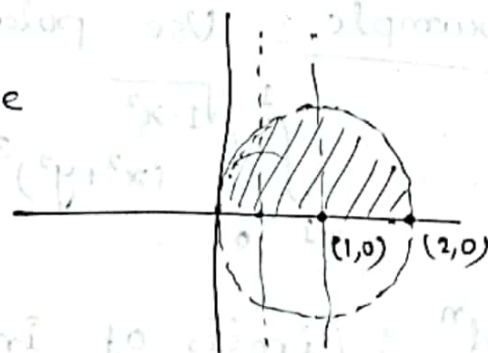
$x=0$ to $x=2$

$y=0$ to $y = \sqrt{2x-x^2}$

$$\Rightarrow x^2 + y^2 - 2x = 0$$

$$\Rightarrow x^2 - 2x + 1 - 1 + y^2 = 0$$

$$\Rightarrow (x-1)^2 + y^2 = 1 \rightarrow \text{circle with center } (1,0), \text{ radius } (1)$$



Now put $x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$ we get.

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx = \int_0^{\pi/2} \int_0^{2\cos\theta} \frac{r \cos \theta}{r} r dr d\theta \quad \left| \begin{array}{l} x^2+y^2-2x=0 \\ \Rightarrow r^2-2r\cos\theta=0 \\ \Rightarrow r=2\cos\theta \end{array} \right.$$

$$= \int_0^{\pi/2} \cos \theta \int_0^{2\cos\theta} r dr d\theta = \int_0^{\pi/2} \cos \theta \left[\frac{r^2}{2} \right]_0^{2\cos\theta} d\theta$$

$$= 2 \int_0^{\pi/2} \cos^2 \theta \cos \theta d\theta = 2 \int_0^{\pi/2} \cos^3 \theta d\theta$$

$$= \frac{2}{4} \int_0^{\pi/2} (\cos 3\theta + 3\cos \theta) d\theta$$

$$= \frac{1}{2} \left[\frac{\sin 3\theta}{3} + 3\sin \theta \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\frac{\sin \frac{3\pi}{2}}{3} + 3\sin \frac{\pi}{2} - 0 \right]$$

$$= \frac{1}{2} \left(-\frac{1}{3} + 3 \right) = +\frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}$$

$$[\cos 3\theta = 4\cos^3 \theta - 3\cos \theta]$$

$$\Rightarrow 4\cos^3 \theta = \cos 3\theta + 3\cos \theta$$

$$\Rightarrow \cos^3 \theta = \frac{\cos 3\theta + 3\cos \theta}{4}$$

Example : Use polar co-ordinates to evaluate

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)^{3/2} dy dx.$$

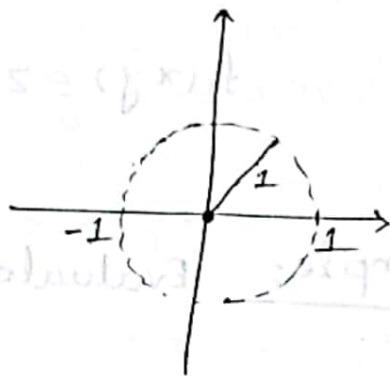
Solⁿ : Limits of integration,

$$x = -1 \text{ to } x = 1$$

$$y=0 \quad \text{to} \quad y=\sqrt{1-x^2}$$

$$\Rightarrow y^2 = 1 - x^2$$

$$\Rightarrow x^2 + y^2 = 1$$



Now put $x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$.

$$\int_0^\pi \int_0^1 r^3 \cdot r dr d\theta = \int_0^\pi \left[\frac{r^5}{5} \right]_0^1 d\theta = \frac{1}{5} \int_0^\pi d\theta$$

$$= \frac{1}{5} \int_0^\pi d\theta = \frac{1}{5} [\theta]_0^\pi = \frac{\pi}{5}$$

Exercise set: 14.3 $\rightarrow$ 1-6, 23-34

Triple integration:

Let $f(x, y, z)$ be a continuous function at every point of a finite region V of three dimensional space. Then triple integral is,

$$\iiint_V f(x, y, z) dV = \int_{x=a}^b \int_{y=g_1(x)}^{g_2(x)} \int_{z=f_1(x,y)}^{f_2(x,y)} f(x, y, z) dz dy dx$$

where V is the region bounded by the curves,

$$a \leq x \leq b$$

$$g_1(x) \leq y \leq g_2(x)$$

$$f_1(x, y) \leq z \leq f_2(x, y)$$

Example; Evaluate $\iiint_R (x+y+z) dx dy dz$ where

$$R: 0 \leq x \leq 1, \quad 1 \leq y \leq 2, \quad 2 \leq z \leq 3.$$

Solution; $\int_0^1 \int_1^2 \int_2^3 (x+y+z) dz dy dx$

$$= \int_0^1 \int_1^2 \left[xz + yz + \frac{z^2}{2} \right]_2^3 dy dx$$

$$= \int_0^1 \int_1^2 \left(3x + 3y + \frac{9}{2} - 2x - 2y - 2 \right) dy dx$$

$$= \int_0^1 \int_1^2 \left(x + y + \frac{5}{2} \right) dy dx$$

$$= \int_0^1 \left[xy + \frac{y^2}{2} + \frac{5}{2}y \right]_1^2 dx$$

$$= \int_0^1 \left(2x + 2 + 5 - xy - \frac{1}{2} - \frac{5}{2} \right) dx$$

$$= \int_0^1 (x + 4) dx$$

$$= \left[\frac{x^2}{2} + 4x \right]_0^1 = \frac{1}{2} + 4 - 0 = \frac{9}{2}$$

(Ans.)

Example: Evaluate, $\iiint_R (x-2y+z) dx dy dz$ where,

$$R: 0 \leq x \leq 1, \quad 0 \leq y \leq x^2, \quad 0 \leq z \leq x+y.$$

Solⁿ:

$$\int_0^1 \int_0^{x^2} \int_0^{x+y} (x-2y+z) dz dy dx$$

$$= \int_0^1 \int_0^{x^2} \left[xz - 2yz + \frac{z^2}{2} \right]_0^{x+y} dy dx$$

$$= \int_0^1 \int_0^{x^2} \left(x(x+y) - 2y(x+y) + \frac{(x+y)^2}{2} \right) dy dx$$

$$= \int_0^1 \int_0^{x^2} \left(x^2 + xy - 2xy - 2y^2 + \frac{x^2}{2} + \frac{y^2}{2} + xy \right) dy dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{3x^2}{2} - \frac{3y^2}{2} \right) dy dx$$

$$= \int_0^1 \left(\frac{3x^2}{2} y - \frac{3}{2} \frac{y^3}{3} \right)_0^{x^2} dx$$

$$= \int_0^1 \left(\frac{3x^2 \cdot x^2}{2} - \frac{(x^2)^3}{2} \right) dx$$

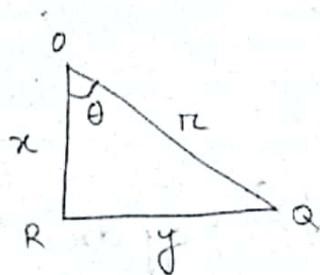
$$= \int_0^1 \left(\frac{3x^4}{2} - \frac{x^6}{2} \right) dx$$

$$= \left[\frac{3}{2} \frac{x^5}{5} - \frac{x^7}{14} \right]_0^1 = \frac{8}{35}$$

Example: $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz \, dy \, dx \longrightarrow \frac{1}{48}.$

☐ Change of Cartesian Co-ordinates into Cylindrical Co-ordinates:

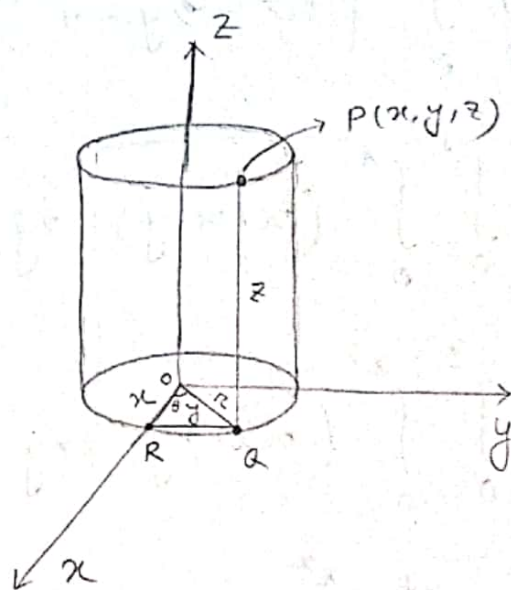
Cartesian $(x, y, z) \longrightarrow$ cylindrical (r, θ, z)



$$\cos \theta = \frac{x}{r} \Rightarrow x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z, \text{ height}$$



So, $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$. $z = z$

and $dx \, dy \, dz = J \, dr \, d\theta \, dz$.

where $J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{vmatrix}$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

$$\Rightarrow dx dy dz = r dr d\theta dz.$$

$$x^2 + y^2 = r^2$$

Hence,
$$\iiint_V f(x, y, z) dx dy dz = \iiint_V f(r \cos \theta, r \sin \theta, z) r dr d\theta dz$$

Example: Evaluate $\iiint_V z(x^2 + y^2 + z^2) dx dy dz$, over the volume of the cylinder $x^2 + y^2 = a^2$ intercepted by the planes $z=0$ and $z=h$.

Solution: Put $x = r \cos \theta$
 $y = r \sin \theta$
 $z = z$

and $dx dy dz = r dr d\theta dz$ in the given integration.

$$\int_0^h \int_0^{2\pi} \int_0^a z(r^2 + z^2) r dr d\theta dz$$

$$= \int_0^h \int_0^{2\pi} \int_0^a (zr^3 + z^3 r) dr d\theta dz$$

$$= \int_0^h \int_0^{2\pi} \left(z \frac{r^4}{4} + z^3 \frac{r^2}{2} \right) \Big|_0^a d\theta dz$$

$$= \int_0^h \int_0^{2\pi} \left(\frac{a^4}{4} z + \frac{a^2}{2} z^3 \right) d\theta dz$$

$$= \int_0^h \left(\frac{a^4}{4} z + \frac{a^2}{2} z^3 \right) [\theta]_0^{2\pi} dz$$

$$= \int_0^h 2\pi \left(\frac{a^4}{4} z + \frac{a^2}{2} z^3 \right) dz$$

$$= 2\pi \left[\frac{a^4}{4} \frac{z^2}{2} + \frac{a^2}{2} \frac{z^4}{4} \right]_0^h$$

$$= 2\pi \left(\frac{a^4}{4} \frac{h^2}{2} + \frac{a^2}{2} \frac{h^4}{4} \right)$$

$$= \left(\frac{a^4 h^2}{4} + \frac{a^2 h^4}{4} \right) \pi$$

$$= \frac{\pi}{4} a^2 h^2 (a^2 + h^2) \quad (\text{Ans.})$$

Example: Evaluate $\iiint z(x^2+y^2) dx dy dz$ over the volume of the cylinder $x^2+y^2=1$ intercepted by the planes $z=2$ and $z=3$.

Solⁿ: $\int_2^3 \int_0^{2\pi} \int_0^1 z(x^2+y^2) r dr d\theta dz$



$$= \int_2^3 \int_0^{2\pi} \int_0^1 z r^3 dr d\theta dz = \int_2^3 \int_0^{2\pi} \left[z \frac{r^4}{4} \right]_0^1 d\theta dz$$

$$= \frac{1}{4} \int_2^3 \int_0^{2\pi} z d\theta dz = \frac{1}{4} \int_2^3 [z\theta]_0^{2\pi} dz = \frac{1}{4} \times 2\pi \int_2^3 z dz$$

$$= \frac{\pi}{2} \left[\frac{z^2}{2} \right]_2^3 = \frac{\pi}{2} \left(\frac{9}{2} - 2 \right) = \frac{5\pi}{4}$$

Example : Use cylindrical co-ordinates to evaluate.

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 dz dy dx$$

Solⁿ : $z = 0$ to $z = 9 - x^2 - y^2$
 $\Rightarrow z = 9 - r^2$

$$y = -\sqrt{9-x^2} \text{ to } y = \sqrt{9-x^2}$$

$$\Rightarrow x^2 + y^2 = 9$$

$$x = -3 \text{ to } x = 3$$

$$\int_0^{2\pi} \int_0^3 \int_0^{9-r^2} (r^2 \cos^2 \theta) r dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} r^3 \cos^2 \theta dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 \left[z r^3 \cos^2 \theta \right]_0^{9-r^2} dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 (9r^3 - r^5) \cos^2 \theta dr d\theta$$

$$= \int_0^{2\pi} \left[\left(\frac{9r^4}{4} - \frac{r^6}{6} \right) \cos^2 \theta \right]_0^3 d\theta$$

$$= \frac{243}{4} \int_0^{2\pi} \cos^2 \theta d\theta$$

$$= \frac{243}{4} \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta$$

$$= \frac{243\pi}{4}$$

▣ Example: Use triple integration in cylindrical co-ords to find the volume of the solid G that is bounded by the hemisphere $z = \sqrt{25 - x^2 - y^2}$, below by the xy -plane, and laterally by the cylinder $x^2 + y^2 = 9$.

Solⁿ: Put $x = r \cos \theta$, $y = r \sin \theta$, $z = z$.

and $dx dy dz = r dr d\theta dz$ in the given integration,

$$\int_0^{2\pi} \int_0^3 \int_0^{\sqrt{25-r^2}} r dz dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 r [z]_0^{\sqrt{25-r^2}} dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 r \sqrt{25-r^2} dr d\theta$$

$$= \int_0^{2\pi} \int_0^3 \sqrt{u} \left(-\frac{du}{2}\right) d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} \left[\frac{u^{3/2}}{3/2} \right]_0^3 d\theta$$

$$= -\frac{1}{3} \int_0^{2\pi} \left[(25-r^2)^{3/2} \right]_0^3 d\theta$$

$$= \int_0^{2\pi} \frac{61}{3} d\theta = \frac{122}{3} \pi.$$

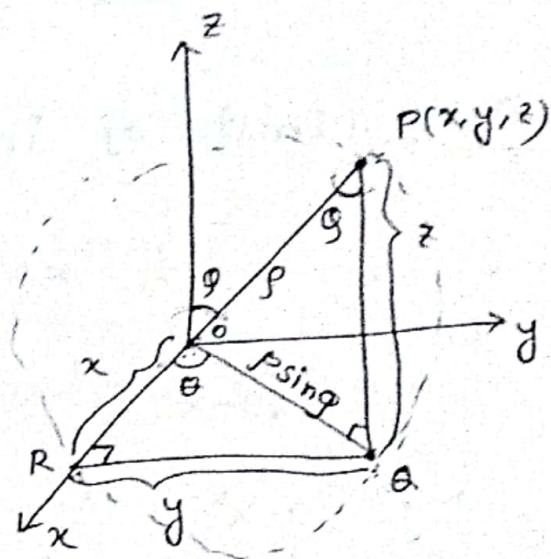
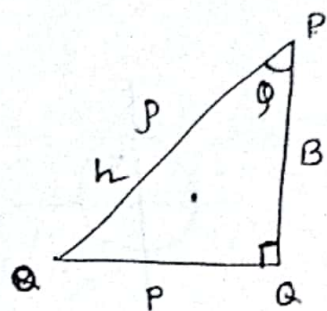
$$\text{Let } u = 25 - r^2$$

$$\Rightarrow du = -2r dr$$

$$\Rightarrow r dr = -\frac{du}{2}$$

Q. Change of cartesian co-ordinates to spherical co-ordinates;

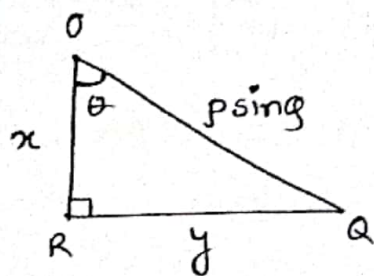
$$OP = \sqrt{x^2 + y^2 + z^2} = \sqrt{p^2} = p$$



$$\sin \phi = \frac{OA}{OP} \Rightarrow OA = p \sin \phi$$

$$\cos \phi = \frac{PA}{OP} \Rightarrow PA = p \cos \phi$$

$$\Rightarrow \boxed{z = p \cos \phi}$$



$$\sin \theta = \frac{y}{p \sin \phi} \Rightarrow \boxed{y = p \sin \phi \sin \theta}$$

$$\cos \theta = \frac{x}{p \sin \phi} \Rightarrow \boxed{x = p \sin \phi \cos \theta}$$

So,
$$\iiint_V f(x, y, z) \, dx \, dy \, dz = \iiint_V f(p \sin \phi \cos \theta, p \sin \phi \sin \theta, p \cos \phi) \underbrace{p^2 \sin \phi}_{J} \, dr \, d\theta \, d\phi$$

Example:
$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} \frac{\sqrt{9-x^2-y^2}}{\sqrt{x^2+y^2+z^2}} dz dy dx$$

Solⁿ: Limits of integration,

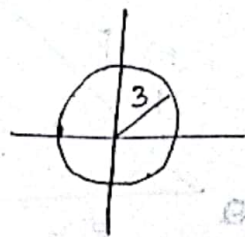
$$x = -3 \text{ to } x = 3$$

$$y = -\sqrt{9-x^2} \text{ to } y = \sqrt{9-x^2}$$

$$\Rightarrow y^2 = 9-x^2$$

$$\Rightarrow x^2+y^2=9$$

$$\Rightarrow x^2+y^2=9$$



$$z = 0 \text{ to } z = \sqrt{9-x^2-y^2}$$

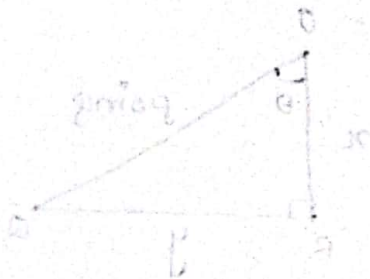
$$\Rightarrow z^2 = 9-x^2-y^2$$

$$\Rightarrow x^2+y^2+z^2=9$$

$$\Rightarrow \rho^2 = 9 = 3^2$$

$$\Rightarrow \rho = 3$$

$$\therefore 0 \leq \rho \leq 3$$



$$0 \leq \theta \leq 2\pi \text{ and } 0 \leq \phi \leq \frac{\pi}{2}$$

Now,
$$\int_0^{\pi/2} \int_0^{2\pi} \int_0^3 \rho \cdot \rho^2 \sin \phi d\rho d\theta d\phi$$

$$= \int_0^{\pi/2} \int_0^{2\pi} \left[\frac{\rho^4}{4} \right]_0^3 \sin \phi d\theta d\phi$$

$$= \frac{81}{4} \int_0^{\pi/2} [\theta]_0^{2\pi} \sin \phi d\phi$$

$$= \frac{81}{2} \pi \left[-\cos \phi \right]_0^{\pi/2}$$

$$= \frac{81\pi}{2} (0 + 1)$$

$$= \frac{81\pi}{2}$$

Example: Using spherical co-ordinates to evaluate

$$\iiint_S e^{(x^2+y^2+z^2)^{3/2}} dV \text{ where } S \text{ is the } \underline{\text{unit sphere}}$$

centered at the origin.

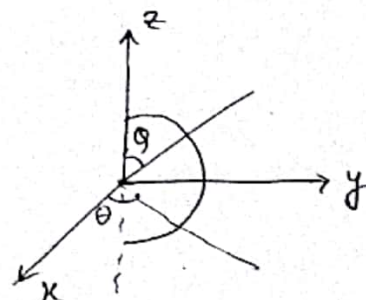
Soln: The limits are: $I = \iiint_S e^{(x^2+y^2+z^2)^{3/2}} dx dy dz$

setting: $x = \rho \sin \phi \cos \theta$

$y = \rho \sin \phi \sin \theta$

$z = \rho \cos \phi$

$dx dy dz = \rho^2 \sin \phi d\rho d\theta d\phi$



The limits are: $0 \leq \rho \leq 1$, $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi$

Now, $\int_0^\pi \int_0^{2\pi} \int_0^1 e^{(\rho^2)^{3/2}} \cdot \rho^2 \sin \phi d\rho d\theta d\phi$

$= \int_0^\pi \int_0^{2\pi} \int_0^1 e^{\rho^3} \cdot \rho^2 \sin \phi d\rho d\theta d\phi$

$= \int_0^\pi \int_0^{2\pi} \int_0^1 e^u \cdot \frac{1}{3} du \sin \phi d\theta d\phi$

Let $\rho^3 = u$

$\Rightarrow 3\rho^2 d\rho = du$

$\Rightarrow \rho^2 d\rho = \frac{1}{3} du$

ρ	0	1
u	0	1

$$= \frac{1}{3} \int_0^{\pi} \int_0^{2\pi} \left[e^u \right]_0^1 \sin \varphi \, d\theta \, d\varphi$$

$$= \frac{e-1}{3} \int_0^{\pi} \int_0^{2\pi} \sin \varphi \, d\theta \, d\varphi = \frac{e-1}{3} \int_0^{\pi} \sin \varphi \left[\theta \right]_0^{2\pi} d\varphi$$

$$= \frac{(e-1)(2\pi)}{3} \int_0^{\pi} \sin \varphi \, d\varphi = \frac{2\pi}{3}(e-1) \left[-\cos \varphi \right]_0^{\pi}$$

$$= -\frac{2\pi}{3}(e-1)(-1-1)$$

$$= \frac{4\pi(e-1)}{3}$$

Exercise 14.6 $\rightarrow$ 1-4, 9-20.

☐ Introduction to differential equations:

Defⁿ: A differential equation or simply DE, is any equation which contains derivatives, either ordinary derivatives or partial derivatives.

Example: 1) $ay'' + by' + cy = g(t)$

2) $\sin(y) \frac{d^2y}{dx^2} = (1-y) \frac{dy}{dx} + y^2 e^{-5y}$

} ODE

3) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \longrightarrow \text{PDE.}$

Order of DE: The order of the DE is the largest derivative present in the DE.

Degree of DE: This is, the degree of the order of DE.

Example: 1) $\frac{d^2x}{dt^2} + k^2x = 0 \longrightarrow$ second order
first degree

2) $(x^2 + y^2) dx + 2xy dy = 0$

$\Rightarrow (x^2 + y^2) \frac{dx}{dy} + 2xy \frac{dy}{dx} = 0 \longrightarrow$ first order
first degree

3) $y''' - 3y' + 2y = 0 \longrightarrow$ third order
first degree

$$4) \quad x \frac{\partial^2 u}{\partial x^2} + 3x = 0 \longrightarrow \text{order} - 2^{\text{nd}} \\ \text{degree} - 1^{\text{st}}$$

$$5) \quad \sin \left(\frac{d^3 y}{dx^3} \right) = \frac{dy}{dx} + x \longrightarrow \text{order} - 1^{\text{st}} \\ \text{degree} - 1^{\text{st}}$$

$$6) \quad \sqrt{\left(\frac{dy}{dx} \right)^2 + 3y} = \frac{d^2 y}{dx^2}$$

$$\Rightarrow \left(\frac{dy}{dx} \right)^2 + 3y = \left(\frac{d^2 y}{dx^2} \right)^2 \quad [\text{squaring both side}]$$

$$\longrightarrow \text{order} - 2^{\text{nd}} \\ \text{degree} - 2^{\text{nd}}$$

$$7) \quad 3y^2 \left(\frac{dy}{dx} \right)^3 - \frac{d^2 y}{dx^2} = \sin(x^2) \longrightarrow \text{order} - 2^{\text{nd}} \\ \text{degree} - 1^{\text{st}}$$

$$8) \quad \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = y \left(\frac{d^3 y}{dx^3} \right)$$

$$\Rightarrow 1 + \left(\frac{dy}{dx} \right)^2 = y^2 \left(\frac{d^3 y}{dx^3} \right)^2 \longrightarrow \text{order} - 3^{\text{rd}} \\ \text{degree} - 2^{\text{nd}}$$

Example (1) : Show that $y(x) = c_1 \sin 2x + c_2 \cos 2x$, where

c_1 and c_2 are arbitrary constants, a solution of

$$y'' + 4y = 0$$

Solution : $y'(x) = 2c_1 \cos 2x - 2c_2 \sin 2x$

$$y''(x) = -4c_1 \sin 2x - 4c_2 \cos 2x$$

$$\begin{aligned} \text{L.H.S} &= -4c_1 \sin 2x - 4c_2 \cos 2x + \cancel{8c_1 \cos 2x} + \cancel{4c_2 \sin 2x} \\ &+ 4c_1 \sin 2x + 4c_2 \cos 2x \\ &= 0 \end{aligned}$$

$\therefore y(x)$ is the solution of $y'' + 4y = 0$. [Shown].

Example: $y'' - 2y' + y = 0$; $y = xe^x$

verify that the indicated function is a solution of the given differential equation.

Solⁿ: $y' = xe^x + e^x$ $y'' = xe^x + 2e^x$

$$\begin{aligned} \text{L.H.S} &= xe^x + 2e^x - 2xe^x - 2e^x + xe^x \\ &= 0 \\ &= \text{R.H.S.} \end{aligned}$$

Trivial Solution: A solution of differential equation that is identically zero on an interval is said to be a trivial solution.

▣ Classification of DE :

- i) Classification by order $\rightarrow$ previous part.
- ii) Classification by Linearity $\rightarrow$
Linear and Nonlinear

An n th order ODE is linear if :

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x) y = f(x)$$

a) The dependent variable y and all its derivatives

$y', y'', \dots, y^{(n)}$ are of the "first degree", that is the power of each term involving y is 1.

(b) The coefficients $a_0, a_1, \dots, a_n$ of $y', y'', \dots, y^{(n)}$ depend at most on the independent variable x .

iii) Classification by Homogeneity:

If $f(x) = 0 \rightarrow$ homogeneous.

$f(x) \neq 0 \rightarrow$ non-homogeneous.

Example: 1. $(1-x)y'' + 3y' = x$

$\Rightarrow (1-x) \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} = x \rightarrow$ Linear, non homogeneous.

2. $y' + \sin y = 0 \rightarrow$ non-linear, homogeneous.

3. $y''' + 3y' + 2y = x^2 \rightarrow$ linear, non-homogeneous

4. $y''' + 4y' + 2y = e^y \rightarrow$ non linear

5. $(1-y) \frac{dy}{dx} + 2y = e^x \rightarrow$ non linear,

6. $\frac{d^4y}{dx^4} + y^2 = 0 \rightarrow$ non linear

7. $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$\Rightarrow 1 + \left(\frac{dy}{dx}\right)^2 = \left(\frac{d^2y}{dx^2}\right)^2 \rightarrow$ non linear.

Exercise 1.1 $\rightarrow$ 1-13. (Zill)

▣ Method of Separation of variables:

Separable equation: A first-order differential eqⁿ of the form, $\frac{dy}{dx} = g(x)h(y)$

is said to be separable or to have separable variables.

Solving a separable DE:

$$\frac{1}{h(y)} \frac{dy}{dx} = g(x)$$

$$\Rightarrow \frac{1}{h(y)} \frac{dy}{dx} dx = g(x) dx$$

$$\Rightarrow \frac{1}{h(y)} dy = g(x) dx$$

$$\Rightarrow \int \frac{1}{h(y)} dy = \int g(x) dx$$

Example: $\frac{dy}{dt} = ky$

$$\Rightarrow \frac{1}{y} dy = k dt$$

$$\Rightarrow \int \frac{1}{y} dy = k \int dt$$

$$\Rightarrow \ln|y| = kt + c$$

$$\Rightarrow y = e^{kt+c} = e^c e^{kt}$$

$$\therefore \boxed{y = e^c e^{kt}} \rightarrow \text{general solution.}$$

Example (2) :
$$\left. \begin{aligned} y \frac{dy}{dx} &= 2x(y+1) \\ y(1) &= 0 \end{aligned} \right\} \text{Initial value problem}$$

Solⁿ : Using separation of variable,

$$\frac{y}{y+1} dy = 2x dx$$

$$\Rightarrow \int \frac{y}{y+1} dy = 2 \int x dx$$

$$\Rightarrow \int \frac{y+1-1}{y+1} dy = 2 \frac{x^2}{2} + c$$

$$\Rightarrow \int \left(1 - \frac{1}{y+1}\right) dy = x^2 + c$$

$$\Rightarrow y - \ln(y+1) + c = x^2 + c$$

$$\Rightarrow x^2 = y - \ln(y+1) + c \longrightarrow \text{general sol}^n$$

when $x=1$, $y=0$ so,

$$1 = 0 + c \Rightarrow c = 1$$

$$\therefore \boxed{x^2 = y - \ln(y+1) + 1} \longrightarrow \text{particular solution.}$$

Example (3) :
$$\left. \begin{aligned} 2xyy' &= 1+y^2 \\ y(2) &= 3 \end{aligned} \right\} \text{IVP}$$

Solⁿ :
$$2xy \frac{dy}{dx} = 1+y^2$$

$$\Rightarrow 2xy dy = (1+y^2) dx$$

$$\Rightarrow \frac{2y}{1+y^2} dy = \frac{1}{x} dx$$

$$\Rightarrow \int \frac{2y}{1+y^2} dy = \int \frac{1}{x} dx$$

$$\Rightarrow \ln(1+y^2) = \ln|x| + c$$

$$\Rightarrow 1+y^2 = e^{\ln|x|+c} = e^{\ln|x|} \cdot e^c = c_1 x$$

$$\Rightarrow y^2 + 1 = c_1 x \quad \text{--- general solution}$$

$$\text{at } x=2 \quad y=3. \quad \text{So, } 9+1 = c_1 \times 2$$

$$\Rightarrow c_1 = 5$$

$$\therefore \boxed{y^2 + 1 = 5x} \quad \text{--- particular solution.}$$

Exercise 2.2 $\rightarrow$ 1-18, 23-26, 31, 32.

Def Linear equation: A first order differential eqn of the form,

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

is said to be a linear equation in the variable y .

$$\text{Standard form: } \frac{dy}{dx} + P(x)y = f(x)$$

Here, Integrating factor, $IF = e^{\int P(x)dx}$

Steps: 1) Put a linear equation into the standard form.

ii) Identify $P(x)$ and find the integrating factor $e^{\int P(x)dx}$.

iii) Multiply both sides of the equation by IF.

The left hand side will be,

$$\frac{d}{dx} \left[e^{\int P(x)dx} \cdot y \right] = e^{\int P(x)dx} \cdot f(x)$$

iv) Integrate both sides, and solve for y .

Example: $\frac{dy}{dx} - 3y = 0$

Here, $P(x) = -3$

$$IF = e^{\int -3dx} = e^{-3x}$$

So, $e^{-3x} \frac{dy}{dx} - 3e^{-3x} y = e^{-3x} \cdot 0$

$$\Rightarrow \frac{d}{dx} (e^{-3x} y) = 0$$

$$\Rightarrow e^{-3x} y = c$$

$$\Rightarrow \boxed{y = c e^{3x}}$$

Example : $x \frac{dy}{dx} - 4y = x^6 e^x$

Solⁿ : $\frac{dy}{dx} - \frac{4}{x} y = x^5 e^x$

Here, $P(x) = -\frac{4}{x}$ and $f(x) = x^5 e^x$

Integrating factor, $I.f = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x} = e^{\ln x^{-4}} = x^{-4}$

So, $x^{-4} \frac{dy}{dx} - 4x^{-5} y = x e^x$

$$\Rightarrow \frac{d}{dx} (x^{-4} y) = x e^x \Rightarrow \int \frac{d}{dx} (x^{-4} y) = \int x e^x$$

$$\Rightarrow x^{-4} y = x \int e^x dx - \int \left\{ \frac{d}{dx} (e^x) \int e^x dx \right\} dx$$
$$= x e^x - e^x + c$$

$$\Rightarrow y = x^5 e^x - x^4 e^x + c_1$$

$$\therefore y(x) = x^5 e^x - x^4 e^x + c_1$$

Exercise {2,3} $\rightarrow$ 1-36.

□ Exact differential equation: A DE is said to be exact if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

$$\downarrow$$
$$\boxed{Mdx + Ndy = 0}$$

For solving the exact DE, we need the following formula,

$$\frac{\partial F}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial F}{\partial y} = N(x, y)$$

where F is the solution to an exact DE.

Example: Solve $2xy dx + (x^2 - 1) dy = 0$

Solⁿ: Here, $M(x, y) = 2xy$

$$N(x, y) = (x^2 - 1)$$

$$\frac{\partial M}{\partial y} = 2x$$

$$\frac{\partial N}{\partial x} = 2x$$

$\therefore$ exact.

There exists a function $f(x, y)$ such that,

$$\frac{\partial f}{\partial x} = 2xy$$

$$\Rightarrow \partial f = 2xy \partial x$$

$$\Rightarrow f(x, y) = \frac{x^2}{2} y + g(y)$$

$$\text{and } \frac{\partial f}{\partial y} = x^2 - 1$$

$$\Rightarrow x^2 + g'(y) = x^2 - 1$$

$$\Rightarrow g'(y) = -1$$

$$\Rightarrow g(y) = -y$$

$$\therefore f(x, y) = x^2 y - y \quad \therefore x^2 y - y = c \text{ is the sol}^n \text{ of the DE.}$$

Example: Solve, $(e^{2y} - y \cos xy) dx + (2xe^{2y} - x \cos xy + 2y) dy = 0$

Solⁿ: Here $M(x, y) = e^{2y} - y \cos xy$
 $N(x, y) = 2xe^{2y} - x \cos xy + 2y$

$$\begin{aligned} \frac{\partial M}{\partial y} &= 2e^{2y} - y \sin xy \cdot (x) - \cos xy \\ &= 2e^{2y} + xy \sin xy - \cos xy \end{aligned}$$

$$\frac{\partial N}{\partial x} = 2e^{2y} - \cos xy + xy \sin xy$$

as, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

Hence, a function $f(x, y)$ exists for which.

$$\frac{\partial f}{\partial x} = M(x, y)$$

$$\Rightarrow \frac{\partial f}{\partial x} = e^{2y} - y \cos xy$$

$$\Rightarrow f(x, y) = x e^{2y} - \sin xy + g(y)$$

$$\text{and } \frac{\partial f}{\partial y} = N(x, y)$$

$$\Rightarrow 2x e^{2y} - x \cos xy + g'(y) = 2x e^{2y} - x \cos xy + 2y$$

$$\Rightarrow g'(y) = \dots + 2y$$

$$\Rightarrow g(y) = \dots + y^2$$

$$\therefore f(x, y) = x e^{2y} - \sin xy + \dots + y^2$$

$$\therefore x e^{2y} - \sin xy + y^2 = c \text{ is the solution.}$$

Example: $(2x^3 - xy^2 + 2y + 3) dx - (x^2y + 2x) dy = 0$

$$\text{Here, } M(x, y) = 2x^3 - xy^2 + 2y + 3 \quad \frac{\partial M}{\partial y} = -2xy + 2$$

$$N(x, y) = -(x^2y + 2x)$$

$$\begin{aligned} \frac{\partial N}{\partial x} &= -(2xy + 2) \\ &= -2xy - 2 \end{aligned}$$

as $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the DE is exact.

Now, $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial x} = M(x, y)$

$$\Rightarrow \frac{\partial f}{\partial x} = 2x^3 - xy^2 - 2y + 3$$

$$\Rightarrow f(x, y) = 2 \frac{x^4}{4} - \frac{x^2}{2} y^2 - 2xy + 3x + g(y)$$

$$\Rightarrow f(x, y) = \frac{x^4}{2} - \frac{x^2 y^2}{2} - 2xy + 3x + g(y)$$

and $\frac{\partial f}{\partial y} = N(x, y)$

$$\Rightarrow -\frac{2x^2 y}{2} - 2x + g'(y) = -x^2 y - 2x$$

$$\Rightarrow -x^2 y - 2x + g'(y) = -x^2 y - 2x$$

$$\Rightarrow g'(y) = 0$$

$$\Rightarrow g(y) = 0$$

$$\therefore C = \frac{x^4}{2} - \frac{x^2 y^2}{2} - 2xy + 3x + 0$$

Non exact differential equations; $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

① If $M(x,y)dx + N(x,y)dy = 0$ is not exact and

$$\mu = \frac{M_y - N_x}{N} = P(x), \text{ } P \text{ is a function of } x \text{ only, then}$$

this equation has an integrating factor,

$$IF = e^{\int P(x) dx}$$

$$= e^{\int \frac{M_y - N_x}{N} dx}$$

$$\text{II) If } \mu = \frac{N_x - M_y}{M} = Q(y) \text{ then } IF = e^{\int \frac{N_x - M_y}{M} dy}$$

Example: Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$

Solⁿ : $M(x,y) = y^4 + 2y$

$$N(x,y) = xy^3 + 2y^4 - 4x$$

$$\frac{\partial M}{\partial y} = 4y^3 + 2$$

$$\frac{\partial N}{\partial x} = y^3 - 4$$

Here $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, the given dE is not exact.

$$\text{Now, } \frac{M_y - N_x}{N} = \frac{4y^3 + 2 - y^3 + 4}{xy^3 + 2y^4 - 4x} \rightarrow \text{not a function of } x \text{ or } y \text{ only.}$$

$$\text{So, } \frac{N_x - M_y}{M} = \frac{y^3 - 4 - 4y^3 - 2}{y^4 + 2y} = \frac{-3y^3 - 6}{y(y^3 + 2)} = \frac{-3(y^3 + 2)}{y(y^3 + 2)} = -\frac{3}{y}$$

which is function of y only.

$$\therefore IF = e^{\int -\frac{3}{y} dy} = e^{-3 \ln y} = e^{-\ln y^3} = y^{-3}$$

$$\therefore y^{-3}(y^4 + 2y) dx + y^{-3}(xy^3 + 2y^4 - 4x) dy = 0$$

$$\Rightarrow (y + 2y^{-2}) dx + (x + 2y - 4xy^{-3}) dy = 0$$

This is an exact de.

$$2.4 \rightarrow 1-5, 7, 9, 10, 12, 13, 17-26$$

Ex. Homogeneous Linear equations with constant coefficient:

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1 \frac{dy}{dx} + a_0 y = 0$$

may be written in the form $y = e^{mx}$.

for example, $y'' - y' - 6y = 0$

$$\text{Let } y = e^{mx}$$

$$y' = me^{mx} \quad y'' = m^2 e^{mx}$$

$$\text{Now, } m^2 e^{mx} - me^{mx} - 6e^{mx} = 0$$

$$\Rightarrow (m^2 - m - 6)e^{mx} = 0$$

$$\Rightarrow m^2 - m - 6 = 0 \quad \text{as, } e^{mx} \neq 0$$

This equation is called characteristic equation.
auxiliary eqn

The root is, $m_1 = 3$ and $m_2 = -2$

The general solution, $y = c_1 e^{3x} + c_2 e^{-2x}$

Ex. Constant coefficient ODE:

$$ay'' + by' + cy = 0$$

$$\text{Let } y = e^{mx}$$

Characteristic equation, $am^2 + bm + c = 0$

$$\Rightarrow m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

i) $b^2 - 4ac > 0 \rightarrow$ two real, distinct roots.

The general solution will be, $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$.

ii) $b^2 - 4ac = 0 \rightarrow$ one repeated root.

The general solution will be, $y = c_1 e^{m_1 x} + c_2 x e^{m_2 x}$.

iii) $b^2 - 4ac < 0 \rightarrow$ complex conjugate roots. $m = \alpha \pm i\beta$.

The general solution will be, $y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x$.

Example: $2y'' - 5y' - 3y = 0$

Solⁿ: auxiliary equation, $2m^2 - 5m - 3 = 0$

$$\Rightarrow 2m^2 - 6m + m - 3 = 0$$

$$\Rightarrow 2m(m-3) + 1(m-3) = 0$$

$$\Rightarrow (m-3)(2m+1) = 0$$

$$\Rightarrow m = 3, -\frac{1}{2}$$

So, The general solution, $y = c_1 e^{3x} + c_2 e^{-1/2 x}$.

Example: $y'' + 4y' + 7y = 0$

auxiliary equation, $m^2 + 4m + 7 = 0$

$$\Rightarrow m = \frac{-4 \pm \sqrt{16 - 28}}{2}$$

$$= \frac{-4 \pm \sqrt{12}i}{2} = \frac{-4 \pm 2\sqrt{3}i}{2}$$

$$= -2 \pm \sqrt{3}i$$

$$m_1 = -2 + \sqrt{3}i \quad m_2 = -2 - \sqrt{3}i$$

The general solution, $y = e^{-2x}(c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x)$.

Higher order differential equation:

Example: $y''' + 3y'' - 4y = 0$

auxiliary equation, $m^3 + 3m^2 - 4 = 0$

$$\Rightarrow m^3 - m^2 + 4m^2 - 4m + 4 - 4 = 0$$

$$\Rightarrow m^2(m-1) + 4m(m-1) + 4(m-1) = 0$$

$$\Rightarrow (m-1)(m^2 + 4m + 4) = 0$$

$$\Rightarrow (m-1)(m+2)^2 = 0$$

$$\Rightarrow m = 1, -2, -2$$

The general solution, $y = c_1 e^x + c_2 e^{-2x} + c_3 x e^{-2x}$.

Exercise 4.3 : 1, 3, 5, 9, 11, 13, 17, 23, 25, 29, 30-35,

37, 39.

II] Undetermined coefficients:

To solve a non homogeneous linear differential eqn,

$$a_n y^n + a_{n-1} y^{n-1} + \dots + a_1 y' + a_0 y = g(x)$$

We must do two things:

- * find the complementary function y_c and
- * find any particular solution y_p of the non-homogeneous equation.

Example: $y'' - 2y' - 3y = 3e^{2t}$

solve the homogeneous: auxiliary equation:

$$m^2 - 2m - 3 = 0$$

$$\Rightarrow m_1 = 3, m_2 = -1$$

$$y_c = c_1 e^{3t} + c_2 e^{-t}$$

a solution to non-homogeneous: Let $y_p = Ae^{2t}$

$$y_p' = 2Ae^{2t} \quad y_p'' = 4Ae^{2t}$$

$$\text{So, } 4Ae^{2t} - 4Ae^{2t} - 3Ae^{2t} = 3e^{2t}$$

$$\Rightarrow -A = 1 \quad \therefore A = -1$$

$$\therefore y_p = -e^{2t}$$

The general solution is, $y = y_c + y_p = c_1 e^{3t} + c_2 e^{-t} - e^{2t}$.

Example: $y'' - 2y' - 3y = 3e^{-t}$

$$y_c = c_1 e^{3t} + c_2 e^{-t}$$

Here, $y_p = Ate^{-t}$

$$y_p'' - 2y_p' - 3y_p = (-2Ae^{-t} + Ate^{-t}) - 2(Ae^{-t} - Ate^{-t}) - 3(Ate^{-t}) = 3e^{-t}$$

$$\Rightarrow -4Ae^{-t} = 3e^{-t}$$

$$\Rightarrow A = -\frac{3}{4} \quad \therefore y_p = -\frac{3}{4}te^{-t}$$

$\therefore$ The general solution is: $y = y_c + y_p$

$$\Rightarrow y = c_1 e^{3t} + c_2 e^{-t} - \frac{3}{4}te^{-t}$$

Non-homogeneity
 e^{mt}

$\sin(mt)$ or $\cos(mt)$

Degree n polynomial

Guess of y_p
 Ae^{mt}

$A\sin(mt) + B\cos(mt)$

$A_0 + A_1t + \dots + A_nt^n$

For more : Table 4.4.1 $\rightarrow$ Trial particular solutions.

Example : $y'' + 4y' - 2y = 2x^2 - 3x + 6$ — (1)

For homogeneous part,

$$m^2 + 4m - 2 = 0$$

$$\Rightarrow m_1 = -2 - \sqrt{6} \quad \text{and} \quad m_2 = -2 + \sqrt{6}$$

The complementary solution is,

$$y_c = c_1 e^{-(2+\sqrt{6})x} + c_2 e^{(-2+\sqrt{6})x}$$

For particular solution, let, $y_p = Ax^2 + Bx + C$

$$y_p' = 2Ax + B$$

$$y_p'' = 2A$$

Substituting these values in (1) $\Rightarrow$

$$2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C = 2x^2 - 3x + 6$$

$$\Rightarrow (-2A)x^2 + (8A - 2B)x + 2A + 4B - 2C = 2x^2 - 3x + 6$$

equating the coefficients of like-powers $\Rightarrow$

$$-2A = 2 \Rightarrow \boxed{A = -1}$$

$$8A - 2B = -3 \Rightarrow \boxed{B = -\frac{5}{2}}$$

$$2A + 4B - 2C = 6 \Rightarrow \boxed{C = -9}$$

$$\therefore y_p = -x^2 - \frac{5}{2}x - 9$$

The general solⁿ, $y = c_1 e^{-(2+\sqrt{6})x} + c_2 e^{(-2+\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$.

Exercise:

44 $\rightarrow$ 1, 3, 5, 7, 17.

Modeling with first order differential equation:

1) Growth and decay:

$$\frac{dx}{dt} = kx, \quad x(t_0) = x_0$$

where k is a constant of proportionality.

Example: A culture initially has P_0 number of bacteria, at $t=1$ h the number of bacteria is measured to be $\frac{3}{2} P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

Solution: Let P be the number of bacteria. Then,

$$\frac{dP}{dt} \propto P \quad \text{and} \quad P(0) = P_0$$

$$\Rightarrow \frac{dP}{dt} = kP$$

$$\Rightarrow \frac{dP}{P} = k dt$$

$$\Rightarrow \ln |P| = kt + C$$

$$\Rightarrow P(t) = e^{kt+C} = C_1 e^{kt}$$

$$\text{at } t=0 \quad C_1 = P_0 \quad \therefore P(t) = P_0 e^{kt}.$$

at $t=1$, we have, $\frac{3}{2} P_0 = P_0 e^k$

$$\Rightarrow e^k = \frac{3}{2}$$

$$\Rightarrow k = \ln \frac{3}{2} = 0.4055$$

$$\text{So, } P(t) = P_0 e^{0.4055t}$$

To find the time at which the numbers of bacteria has tripled, we solve,

$$3P_0 = P_0 e^{0.4055t}$$

$$\Rightarrow 0.4055t = \ln 3$$

$$\Rightarrow t = 2.71 \text{ h (approx)}$$

Half life:

The half life is simply the time it takes for one-half of the atoms in an initial amount A_0 to disintegrate, or transmute, into the atoms of another element.

Example: A breeder reactor converts relatively stable uranium-238 into the isotope plutonium-239. After 15 years it is determined that 0.043% of the initial

amount A_0 of plutonium has disintegrated. Find the half life of this isotope if the rate of disintegration is proportional to the amount remaining.

Solution: Let $A(t)$ be the amount of plutonium remaining at time t .

$$\frac{dA}{dt} = kA, \quad A(0) = A_0$$

$$\Rightarrow A(t) = c_1 e^{kt}$$

$$\text{at } t=0 \quad c_1 = A_0 \quad \therefore A(t) = A_0 e^{kt}$$

If 0.043 % of the atoms of A_0 have disintegrated, then 99.957 % of the atoms remains.

$$\text{So, } 0.99957 \times A_0 = A_0 e^{15k}$$

$$\Rightarrow k = \frac{1}{15} \ln 0.99957$$

$$= -0.00002867$$

$$\therefore A(t) = A_0 e^{-0.00002867t}$$

$$\text{The half life is } A(t) = \frac{1}{2} A_0$$

$$\Rightarrow A_0 e^{-0.00002867t} = \frac{1}{2} A_0$$

$$\Rightarrow t = \frac{\ln 2}{0.00002867} \approx 24180 \text{ year.}$$

▣ Newton's law of cooling/warming:

$$\frac{dT}{dt} = k(T - T_m)$$

$k = \text{constant}$

$T(t)$ = Temperature of the object for $t > 0$

T_m = Ambient temperature

Example: A thermometer is removed from a room where the temperature is 70°C and is taken outside, where the air temperature is 10°F . After one-half minute the thermometer reads 50°F . What is the reading of the thermometer to reach 16°F at $t = 1$ min.

Solⁿ: Using Newton's law of cooling,

$$\frac{dT}{dt} = k(T - 10)$$

according to question, $T(0) = 70^\circ\text{F}$ and $T(\frac{1}{2}) = 50^\circ\text{F}$

$$\text{Now, } \frac{dT}{T-10} = k dt$$

$$\Rightarrow \ln(T-10) = kt + c$$

$$\Rightarrow T-10 = c_1 e^{kt} \quad \therefore T(t) = 10 + c_1 e^{kt}$$

at $t=0$, $T(t)=$

$$10 + c_1 e^{k \times 0} = 70^\circ$$

$$\Rightarrow c_1 = 60^\circ$$

and, at $t = \frac{1}{2}$

$$10 + 60 \times e^{k/2} = 50^\circ$$

$$\Rightarrow e^{k/2} = \frac{40}{60} = \frac{2}{3}$$

$$\Rightarrow \frac{k}{2} = \ln\left(\frac{2}{3}\right)$$

$$\Rightarrow k = 2 \ln\left(\frac{2}{3}\right)$$

$$\therefore T(t) = 10 + 60 e^{-0.811t}$$

$$\begin{aligned} \text{at } t=1 \quad T(1) &= 10 + 60 e^{-0.811 \times 1} \\ &= 36.67^\circ \text{ F.} \end{aligned}$$

Exercise 3.1 $\rightarrow$ 1-3, 6, 7, 13-17.